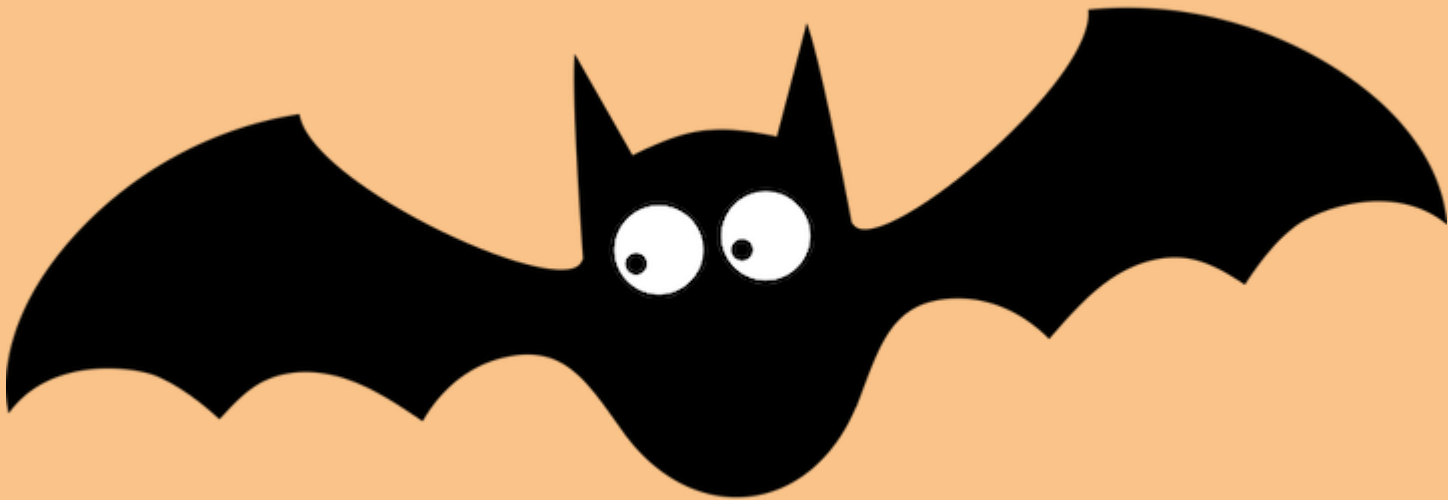




Ensemble Method and Boosting

Son Nguyen

Netflix Prize

- 1 million USD to any one that can improve Netflix's rating algorithm...

Netflix Prize

- 1 million USD to any one that can improve Netflix's rating algorithm **by 10%**

Netflix Prize

- **The winning team:**

Netflix Prize

- **The winning team:** BellKor's Pragmatic Chaos, used **ensemble models**"

Netflix Prize

- **The winning team:** BellKor's Pragmatic Chaos, used **ensemble models**"
- **The second-place team's name** is...

Netflix Prize

- **The winning team:** BellKor's Pragmatic Chaos, used **ensemble models**
- **The second-place team's name** is "The Ensemble"

Netflix Prize

- "During the nearly 3 years of the Netflix competition, there were two main factors which improved the overall accuracy:
 - The quality of the individual algorithms, and
 - the **ensemble idea**"

Ensemble Success

"XGBoost (an ensemble algorithm) is an algorithm that has recently been dominating applied machine learning and Kaggle competitions for structured or tabular data." [Link](#)

- List of machine learning winning solutions with XGBoost :
<https://github.com/dmlc/xgboost/tree/master/demo#machine-learning-challenge-winning-solutions>

Ensemble Success

Adaboost (an ensemble algorithm) won 2003 **Godel Prize**: AdaBoost demonstrated novel possibilities in analysing data and is a permanent contribution to science even beyond computer science. [Link](#)

Ensemble Success

AdaBoost (with decision trees as the weak learners) is often referred to as the best out-of-the-box classifier. [Link](#)

Ensemble Success

Leo Brieman, who invented "Bagging" and "Random Forest" crowned AdaBoost the "best off-the-shelf classifier in the world (2000).

Ensemble Machine Learning Approach

- An ensemble is a composite model, combining a series of low performing **classifiers** (classification models) or **learners** with the aim of creating an improved classifier. weak learners

Strong learner

example: Random forest

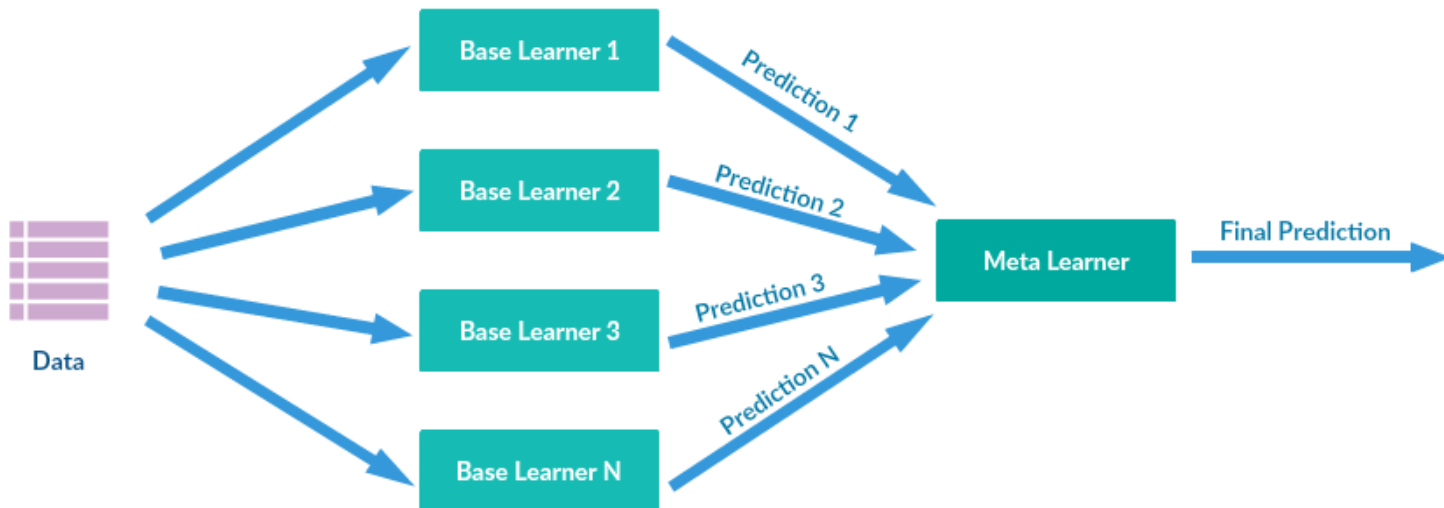
Ensemble Machine Learning Approach

- An ensemble is a composite model, combining a series of low performing **classifiers**(classification models) or **learners** with the aim of creating an improved classifier.
- Three common ensemble:
 - Stacking
 - Bagging
 - Boosting

← random forest

Stacking

- Stacking combines multiple base learners predictions into a new data set.
- This new data are treated as the **input data** for another learner (meta learner).



① data for base learner 1 (M_1)

x_1	x_2	y	M_1
1	2	1	0
2	3	0	0

② data for base learner 2 (M_2)

x_1	x_2	y	M_2
1	2	1	1
2	3	0	1

③ data for base learner 3 (M_3)

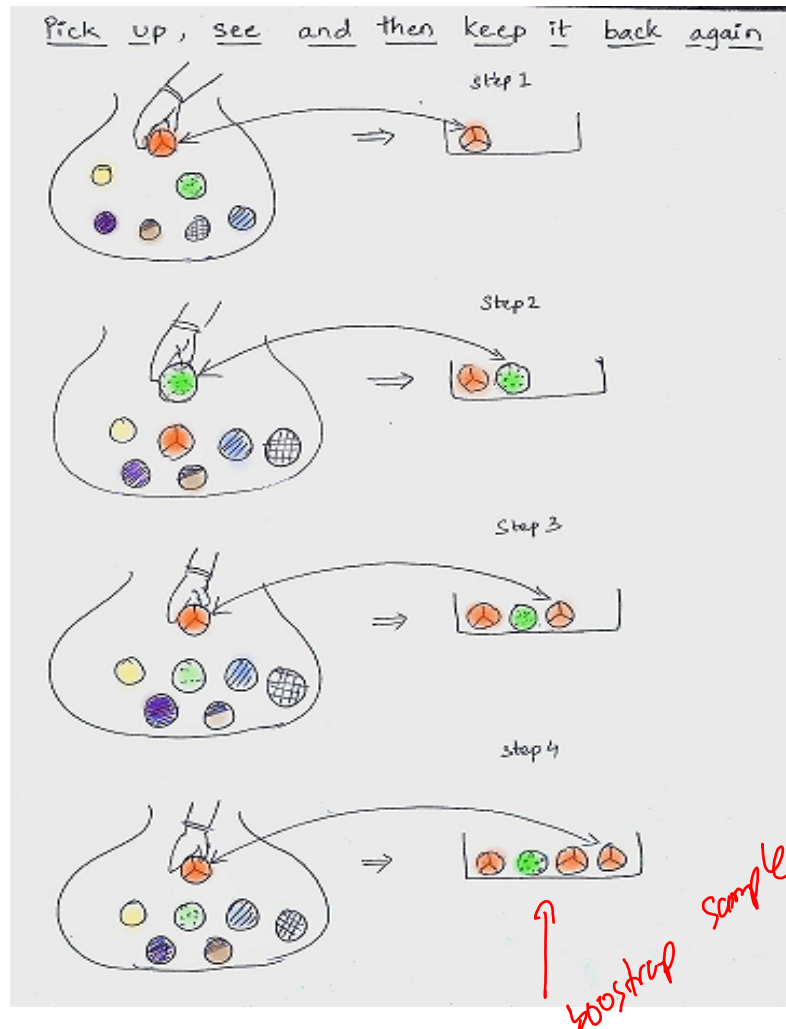
x_1	x_2	y	M_3
1	2	1	1
2	3	0	0

M_1 Prediction	M_2 prediction	M_3 Prediction	y
0	1	1	1
0	1	0	0

Data for meta learner

Bagging = Bootstrap Aggregating

- Step 1: Bootstrapping

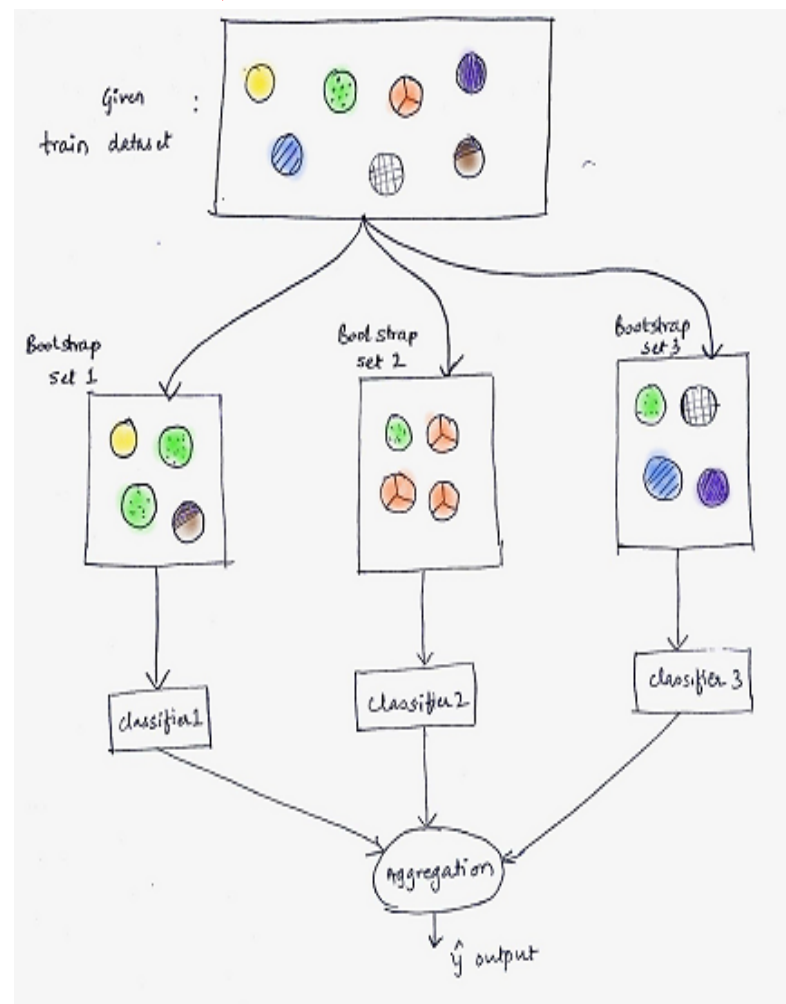


Bagging = Bootstrap Aggregating

Step 1

Step 2

- Step 2: Aggregating



ID	X_1	X_2	Y
1	7	3.5	3.7
2	20	3	4.0
3	2	3.5	2.5
4	30	2.0	3.8

Weak Learner 1

ID	Y	M_1	error
1	3.7	1.0	2.7
2	4.0	3.0	1.0
3	2.5	2.6	- .1
4	3.8	0.5	3.3

data for
 M_2

X_1 : # hours studied / week

X_2 : high school GPA

Y : college GPA

Weak Learner 2 will pay
more attention to the most
wrong prediction of M_1

ID	Y
4	3.8
4	3.8
4	3.8
4	3.8
1	3.7
1	3.7
1	3.7
2	4.0
2	4.0
3	2.6

IO	y	M_2	error $\leftarrow M_2$
1	3.7	3.6	.1
2	4.0	2.5	1.5
3	2.5	3.0	-.5
4	3.8	4.0	-.2

Here M_2 does worse on observations that M_1 did well

but M_2 does well on _____ M_1 did not do well

M_3 comes in to fix the errors of M_2 ,

And so on ... to M_{1000}

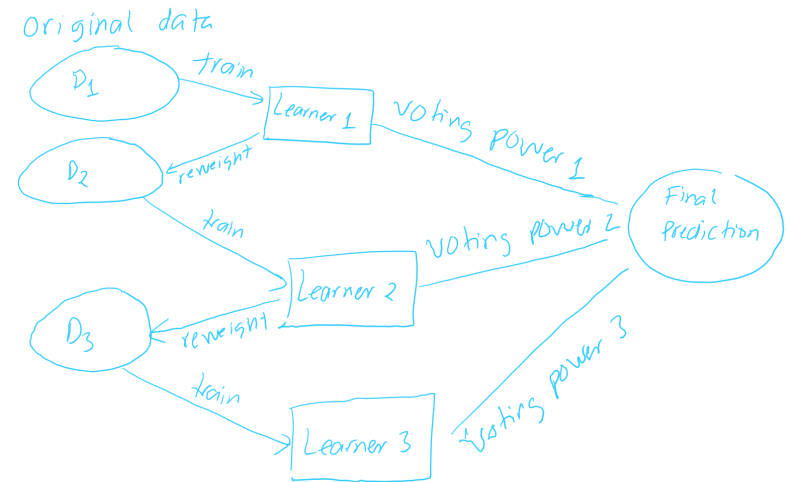
In the end, we combine all $M_1, M_2, \dots, M_{1000}$. Each of these model has a voting power.

who would get the higher voting powers??

models with "less error" will have higher voting powers

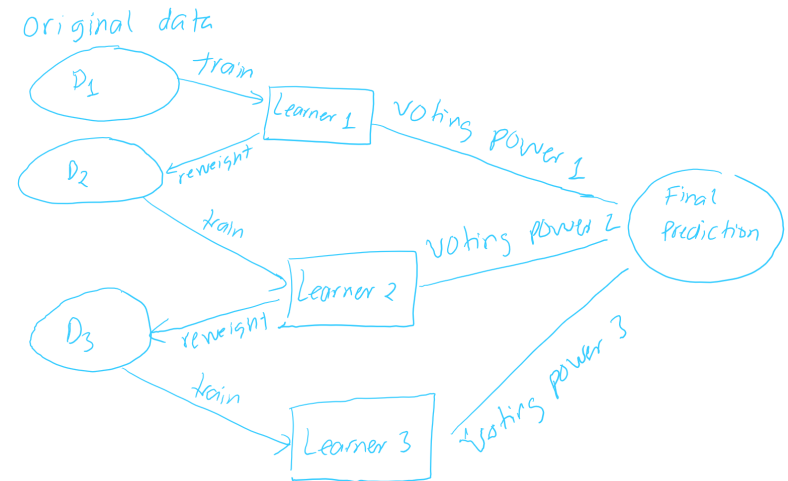
Boosting

- Weak learners are sequentially converted into a strong learner.



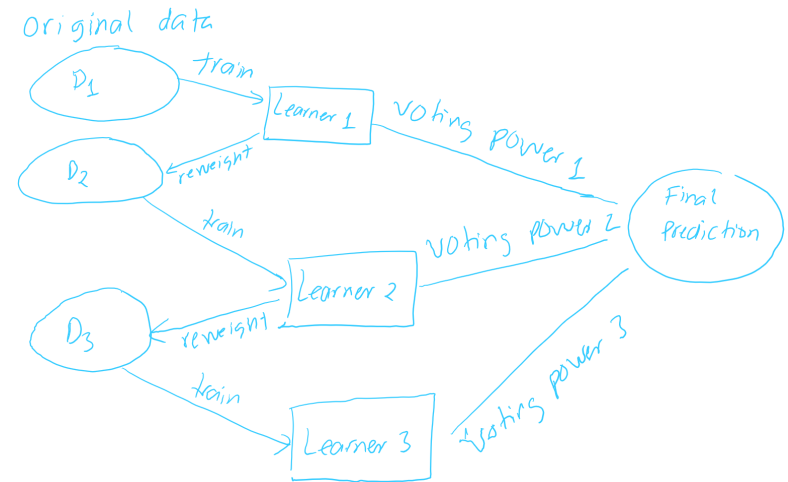
Boosting

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- Each learner in the sequence tries to correct its predecessor.



Boosting

- Weak learners are sequentially converted into a strong learner.
- Each learner in the sequence tries to correct its predecessor.
- Individually, Learner 1, Learner 2 and Learner 3 are **weak learners** (a little better than a coin toss).

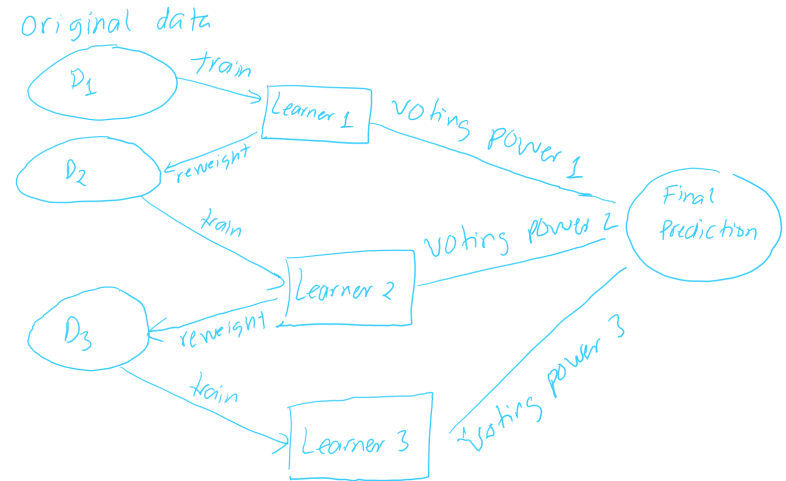


weak learners:

- ① Random Guess
- ② Tree with 1 split
(Stump)
- ③ Predict by the majority / average of the output.

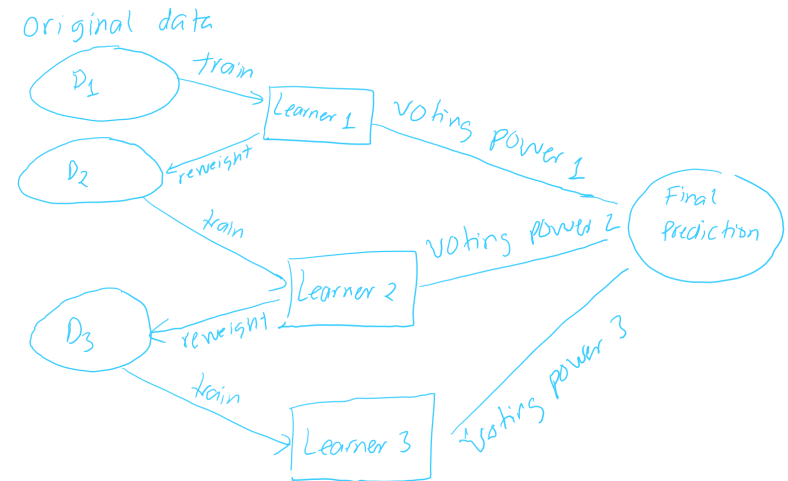
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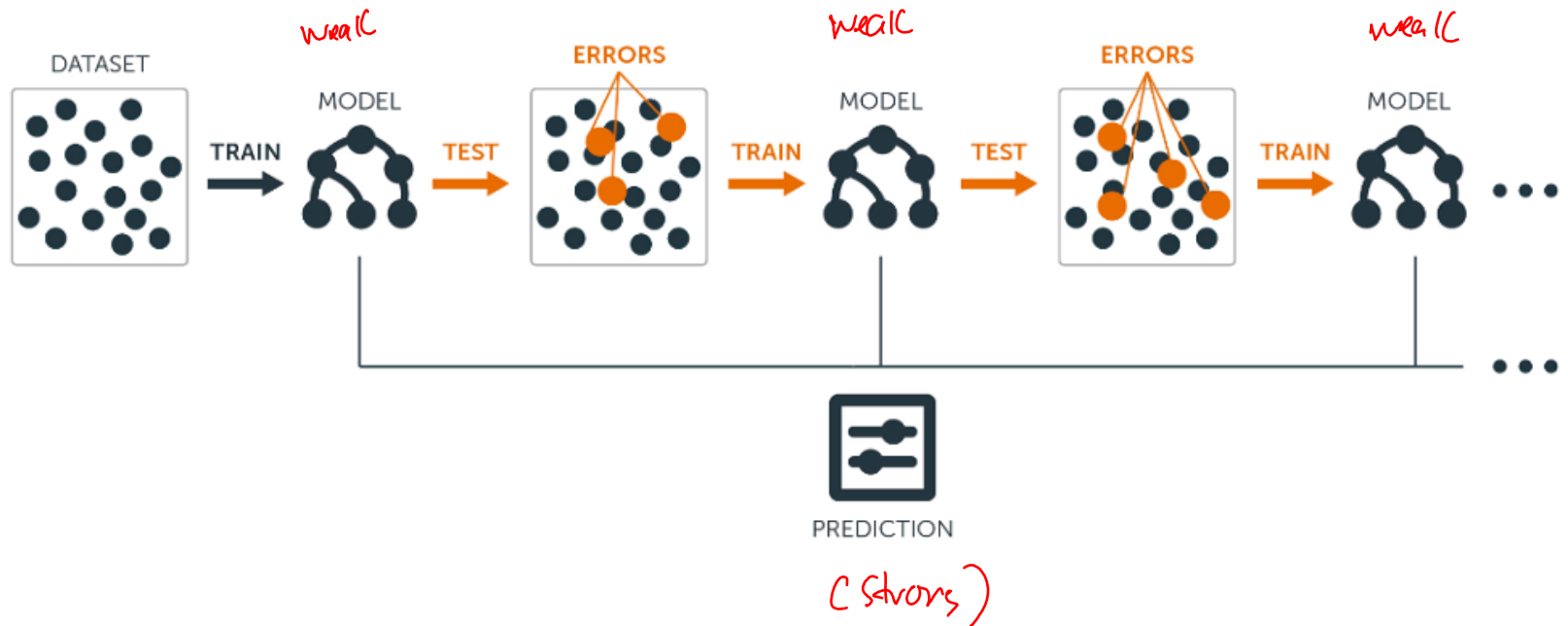


Boosting

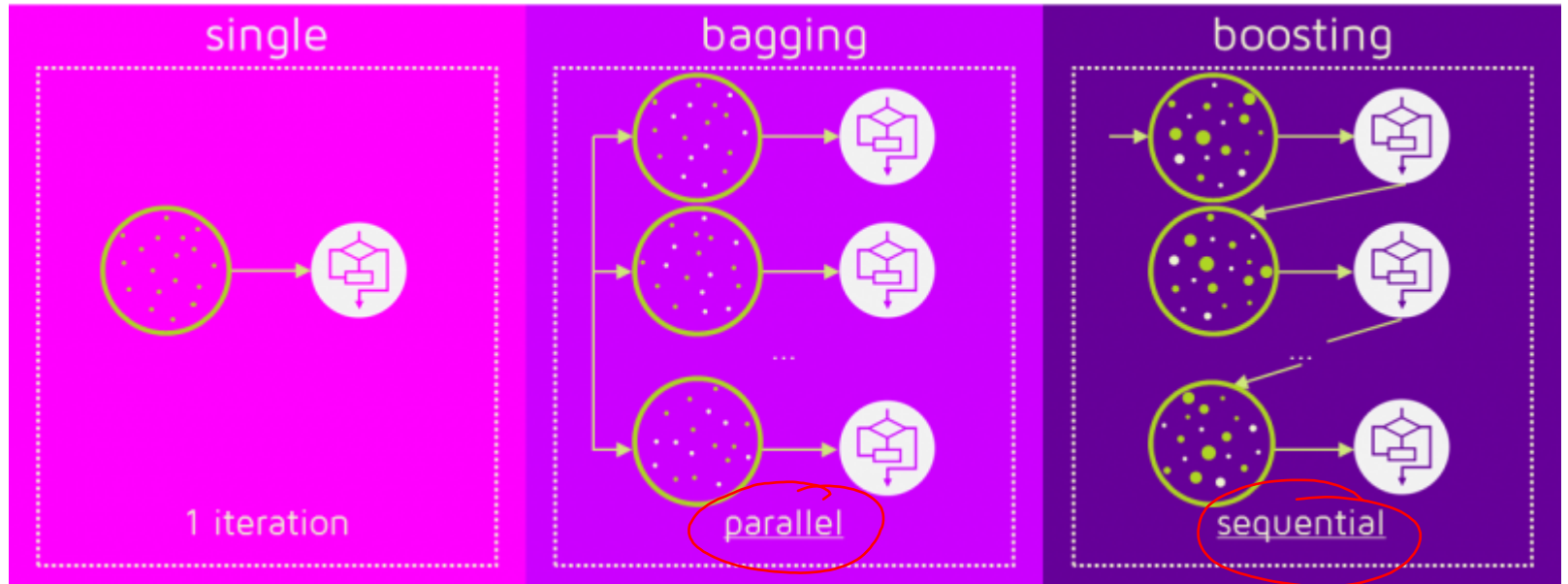
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- Each learner in the sequence tries to correct its predecessor.
- Individually, Learner 1, Learner 2 and Learner 3 are **weak learners** (a little better than a coin toss).
- The learners are usually a tree with 2 leaves (**Stump**).
- In D_2 , the wrong misclassified of Learner 1 gets higher weights.
- In D_3 , the wrong misclassified of Learner 2 gets higher weights.



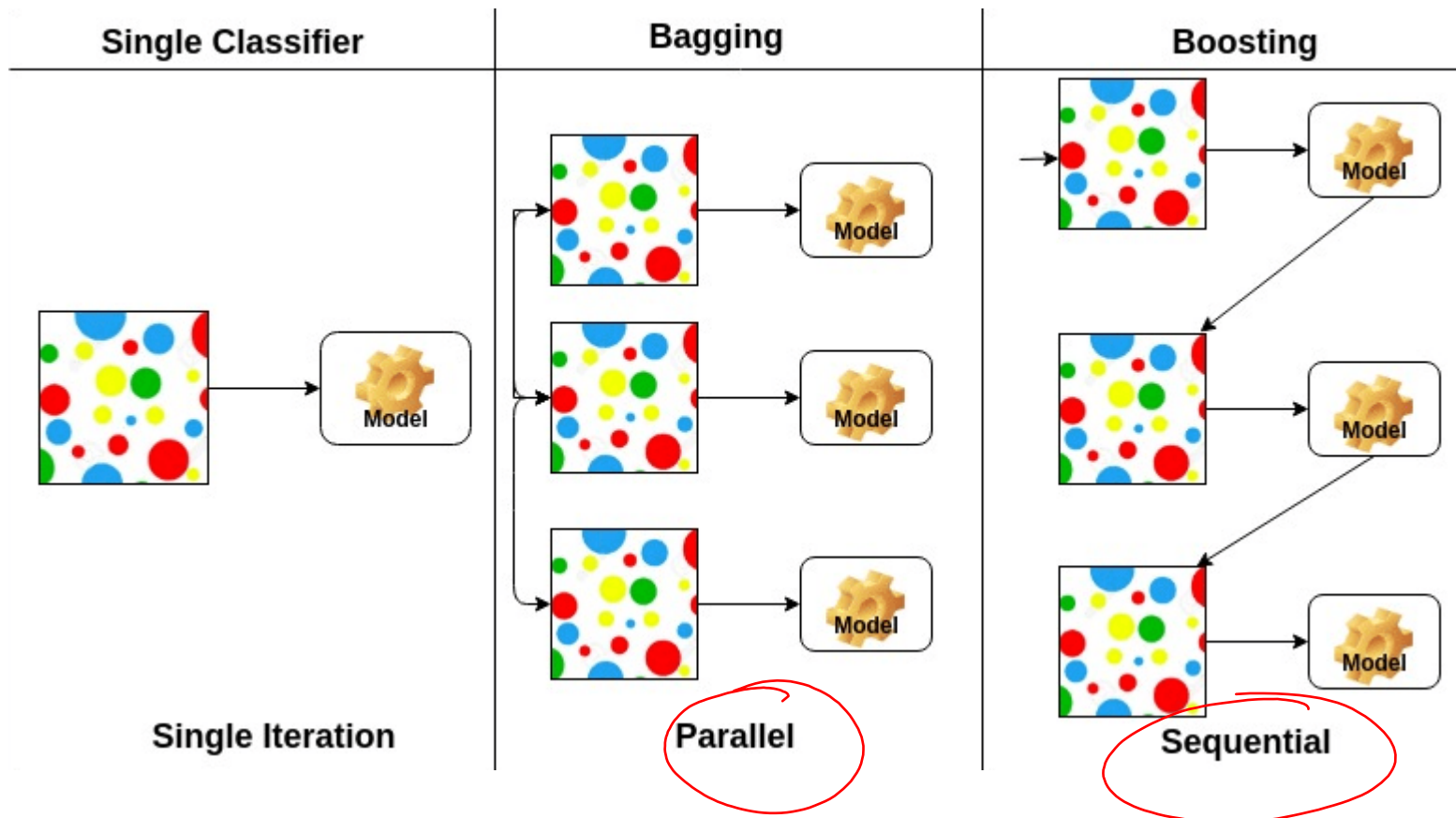
Boosting



Bagging vs. Boosting



Bagging vs. Boosting



Types of Boosting

- ① • Adaboost
- ② • Gradient Boosting
- ③ •
- ④ •
- ⑤ •
- ⋮

Adaboost

Adaboost

Adaboost, Clearly Explained

- Demonstration by StatQuest
- [Link](#)